B.Tech CSE - Third Year | Lab 13

Assignment 13: Recursion on Arrays • *Aim: Process array elements recursively.*

Experiment No. 13.1: Recursive sum of array

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_01_Recursive_sum_of_array {
    static int sum(int[] a, int i) {
        if (i == a.length) return 0;
        return a[i] + sum(a, i + 1);
    }
    public static void main(String[] args) {
        int[] a = {3, 6, 9, 12, 15};
        System.out.println(sum(a, 0));
    }
}
```

3. **Output:** `45`
4. **Viva-Voce:** What is the stopping condition? Why does the index increase? What is the sum of the given array?

Experiment No. 13.2: Find array maximum recursively

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_02_Find_array_maximum_recursively {
    static int max(int[] a, int i) {
        if (i == a.length - 1) return a[i];
        return Math.max(a[i], max(a, i + 1));
    }
    public static void main(String[] args) {
        int[] a = {3, 6, 9, 12, 15};
        System.out.println(max(a, 0));
    }
}
```

3. **Output:** `15`
4. **Viva-Voce:** Why is the last element the base case? What does Math.max return? What is the maximum value here?

Experiment No. 13.3: Print array in reverse recursively

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_03_Print_array_in_reverse_recursively {
    static void printReverse(int[] a, int i) {
        if (i < 0) return;
        System.out.println(a[i]);
        printReverse(a, i - 1);
    }
    public static void main(String[] args) {
        int[] a = {3, 6, 9, 12, 15};
        printReverse(a, a.length - 1);
    }
}
```

3. **Output:**

```
15
12
9
6
3
```

4. **Viva-Voce:** Why start from the last index? What is the base case? How does the index change on each call?